Magnetic Field from a Force Balance, and Dipole Energy (completed)

Goal.

- Apply the velocity-selector force balance to solve for the *field* instead of the speed — same equation, different unknown.
- Use $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$ to find a dipole's potential energy at a given orientation, and the *change* in energy between two different orientations.

1. Worked example: finding the field in a velocity selector

Recall the velocity selector: a charge moving through crossed $\vec{E}$ and $\vec{B}$ fields goes straight only when the electric and magnetic forces exactly cancel, $qE = qvB$. Last time you solved that for the speed v. This time you're given the speed and have to solve for the field.

A charged particle moves with speed $v = 3.0 \times 10^5$ m/s through a region containing a uniform electric field $E = 6.0 \times 10^4$ N/C, directed vertically downward. The particle passes straight through without deflecting. What is the magnitude of the magnetic field $\vec{B}$ present in the region? Assume $\vec{B}$ is perpendicular to both $\vec{v}$ and $\vec{E}$, and the particle is positively charged.

MODEL. The particle goes straight only if the electric force and the magnetic force on it are equal and opposite, exactly canceling.

SOLVE. Set the force magnitudes equal:

$$qE = qvB$$

The charge q cancels — the balance condition doesn't depend on it. Solve for B:

$B = \frac{E}{v}$

Plug in the numbers:

$$B = \frac{6.0 \times 10^4 \text{ N/C}}{3.0 \times 10^5 \text{ m/s}} = 0.20 \text{ T}$$

Direction. The electric force on the positive charge points downward (along $\vec{E}$), so the magnetic force $q\vec{v} \times \vec{B}$ must point upward to cancel it. With $\vec{v}$ horizontal, the right-hand rule then fixes which way $\vec{B}$ must point.

Check. If the particle's speed were doubled but E stayed the same, would the required B go up, go down, or stay the same? Does that match your intuition — does a faster particle need a *stronger* or *weaker* field to still be balanced?

$B = E/v$, so doubling v with E fixed *halves* the required B. That matches intuition: a faster particle already feels a bigger magnetic force at a given B (since $F_B = qvB$ grows with v), so a *weaker* field is enough to match the same electric force.

2. Worked example: energy of a magnetic dipole, two orientations

Recall the energy of a magnetic dipole in a field: $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$, lowest when $\vec{\mu}$ is aligned with $\vec{B}$ ($\theta = 0$) and highest when anti-aligned ($\theta = 180°$).

A magnetic dipole with moment $\mu = 0.60$ $A{\cdot}m^2$ sits in a uniform field $B = 0.30$ T. Someone rotates it from $\theta_1 = 40°$ to $\theta_2 = 110°$ (both measured from $\vec{B}$). What is the change *in the dipole's potential energy?*

MODEL. The energy change is just the energy in the final orientation minus the energy in the initial one: $\Delta U = U_2 - U_1$.

SOLVE. Find the energy at each angle separately:

$$U_1 = -\mu B \cos\theta_1 = -(0.60)(0.30)\cos 40° = -0.138 \text{ J}$$

$$U_2 = -\mu B \cos\theta_2 = -(0.60)(0.30)\cos 110° = +0.0616 \text{ J}$$

Now subtract:

$\Delta U = U_2 - U_1 = 0.0616 - (-0.138)$

Plug in the numbers:

$\Delta U = 0.20$ J

Check. Is ΔU positive or negative here? $\theta_1 = 40°$ is closer to alignment (0°) than $\theta_2 = 110°$ is. Does the sign of ΔU match the fact that torque always twists a dipole *toward* alignment — so moving *away* from alignment should take energy *in*, not give it up?

ΔU is positive: energy went *up*. That's exactly consistent with the torque always pulling $\vec{\mu}$ toward alignment with $\vec{B}$ — rotating the dipole *away* from alignment (here, from 40° to 110°, farther from 0°) works against that torque, which costs energy, just like lifting a pendulum away from its lowest point.